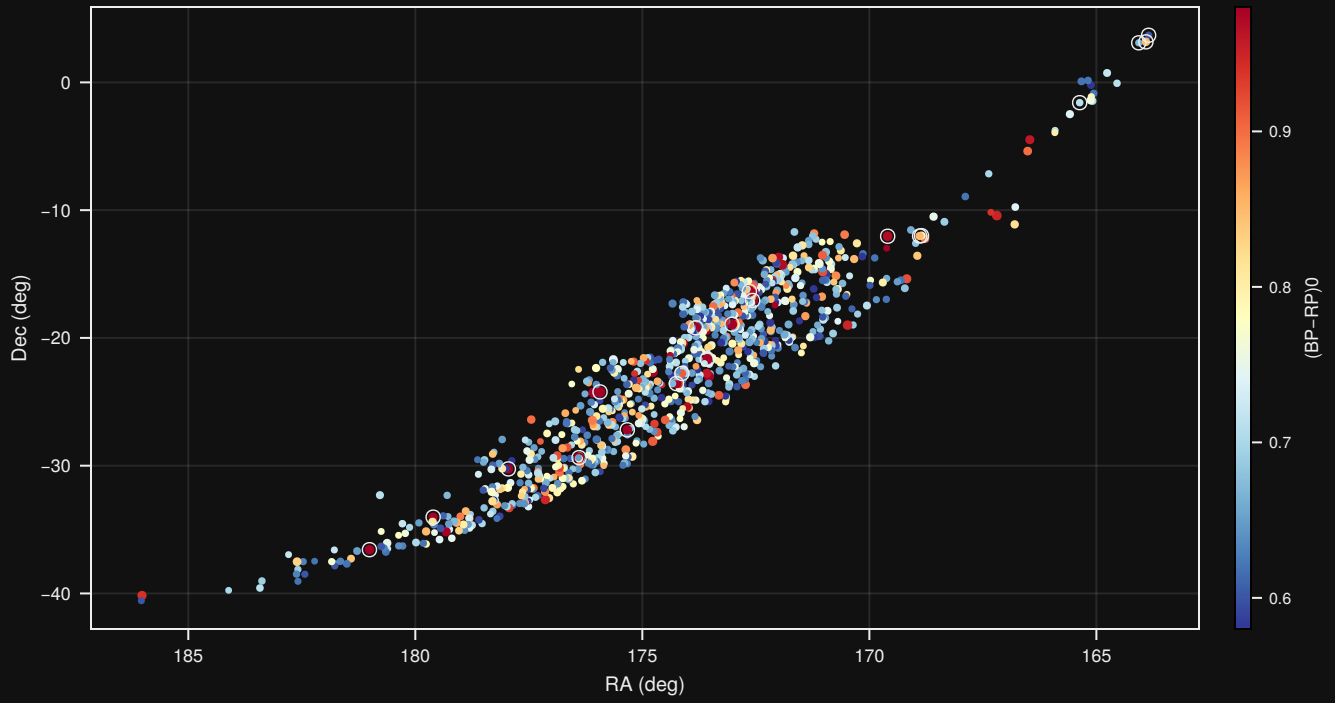
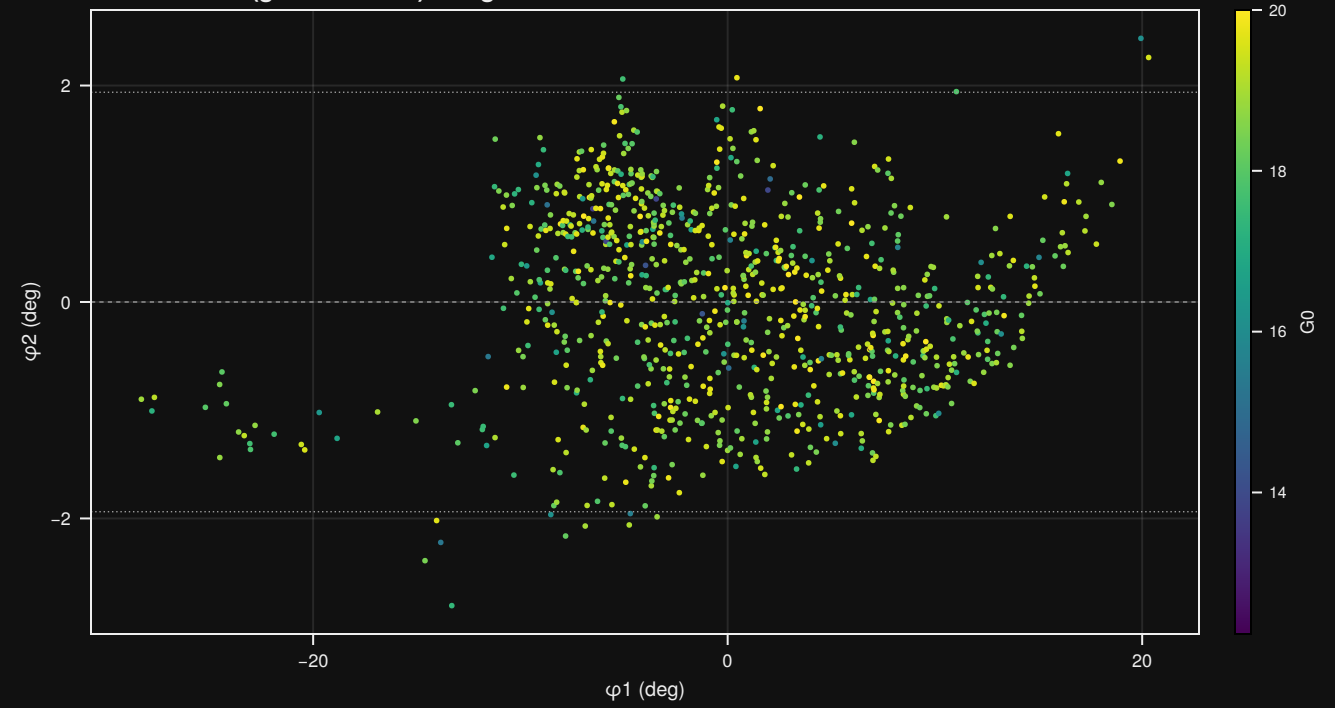
Ylgr

MEMBERSHIP	
Stars in catalog	919
with Gaia GSP-Phot	473 (51%)
with radial velocity	20
Stream id (sID)	41
GEOMETRY · great-circle fit	
Centroid RA, Dec	174.3, -22.1 deg
Centroid l, b	280.7, 37.5 deg
Pole RA, Dec	256.0, 18.5 deg
Length (φ1, 2–98%)	31.8 deg
Width σ(φ2), robust	0.97 deg
KINEMATICS	
pmRA* med ± σ	-0.44 ± 0.43 mas/yr
pmDE med ± σ	-7.56 ± 1.23 mas/yr
μφ1* med ± σ	6.95 ± 1.27 mas/yr
μφ2 med ± σ	-2.99 ± 0.55 mas/yr
Parallax (cat.) med	0.133 ± 0.249 mas
Parallax (wtd, ZP-corr.)	0.131 ± 0.011 mas
Distance	d = 7.6 ± 0.6 kpc
Physical length	4.25 kpc
RV median ± σ(MAD)	323.0 ± 21.3 km/s
RV gradient dV/dφ1	3.87 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.2–20.0 (18.8)
(BP−RP)0 median	0.70
GSP-Phot [M/H] (n=473)	-1.78 ± 0.78
GSP-Phot Teff median	5793 K
RUWE > 1.4	0.0 %
PM clusters (largest)	60 (59 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	919 | 919
n_RV catalog | paper	20 | 20
<[Fe/H]> paper	-2.09
Stellar mass (paper)	11000 Msun
Pole offset vs paper	1.3 deg

Sky map · catalog members (n = 919) · size ∝ brightness · ring = has RV

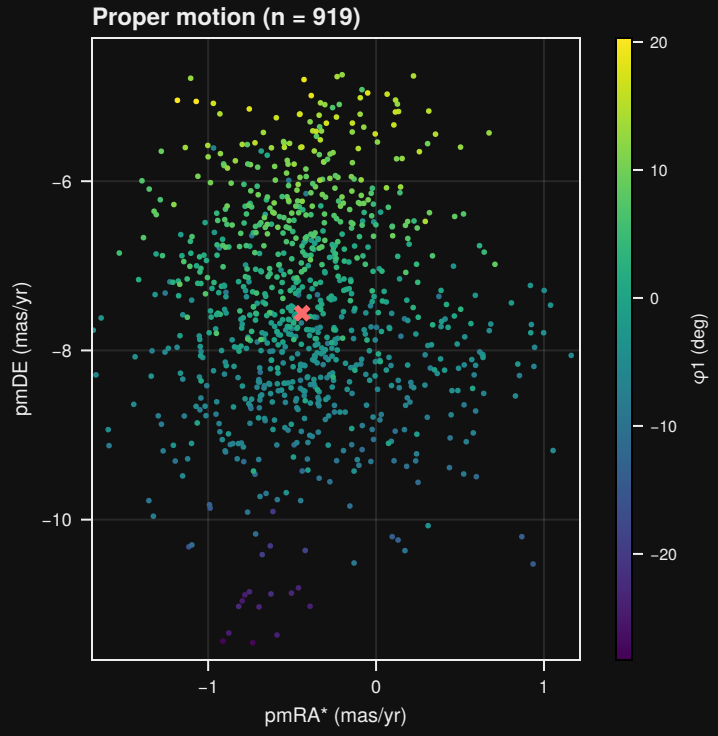


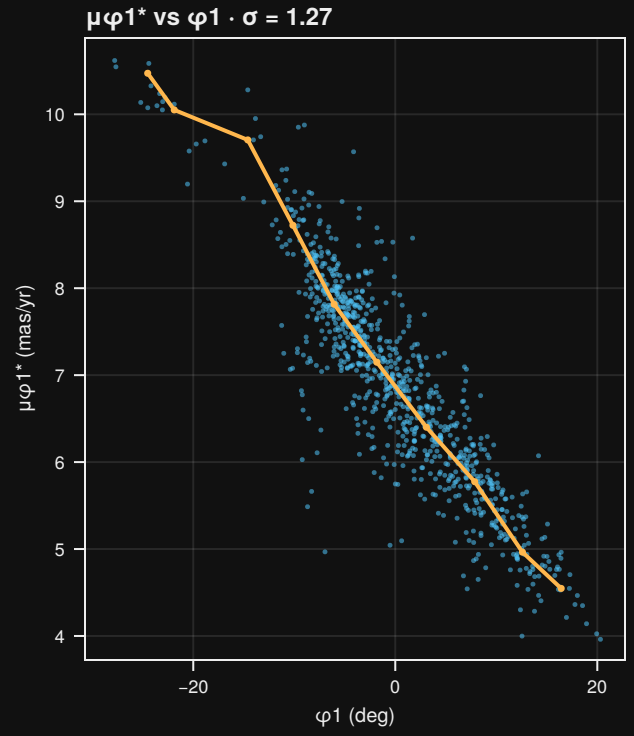
Stream frame (great-circle fit) · length 31.8° · width σ 0.97°

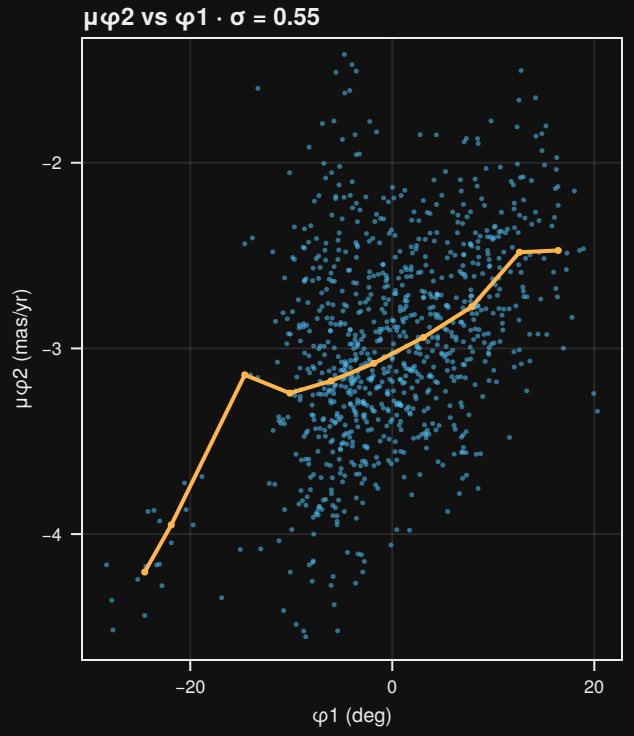


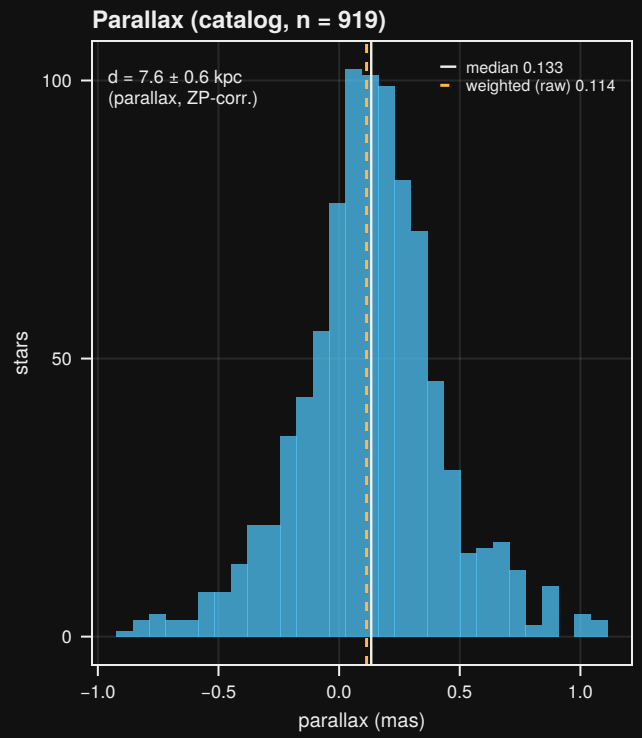
Ylgr

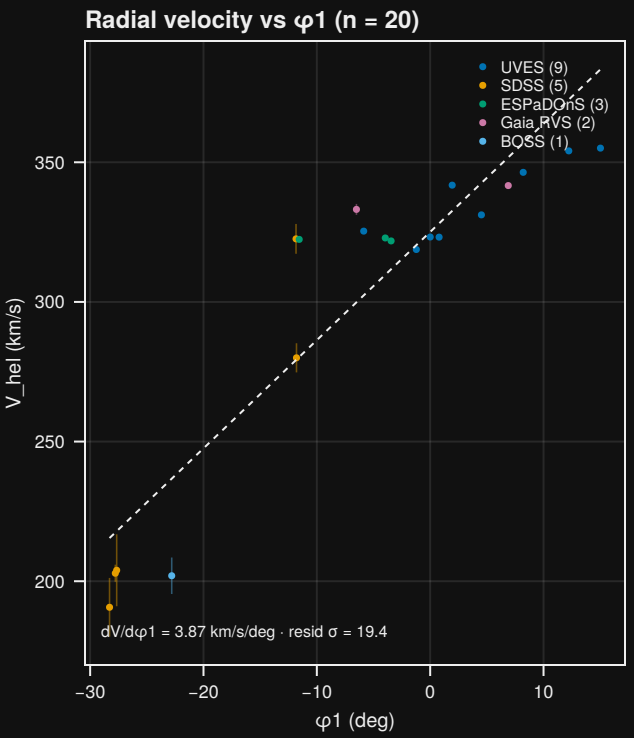
Kinematics · proper motion · parallax · radial velocity

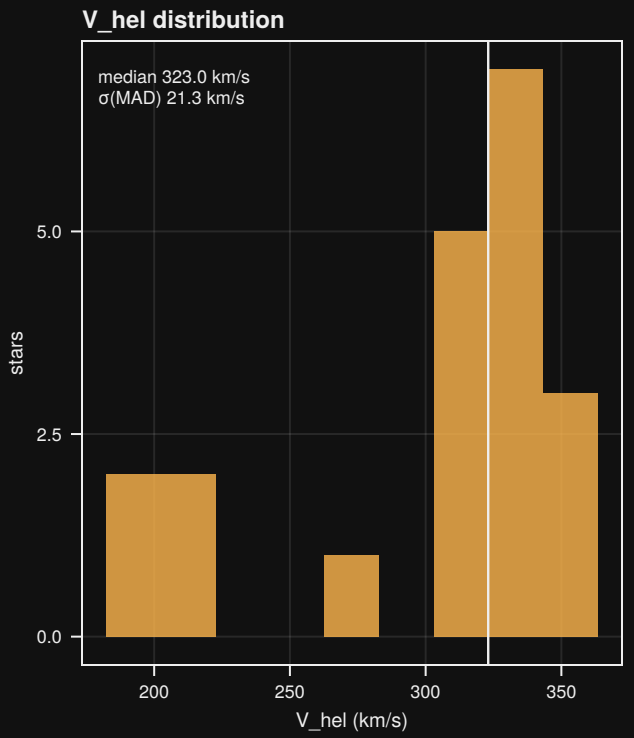






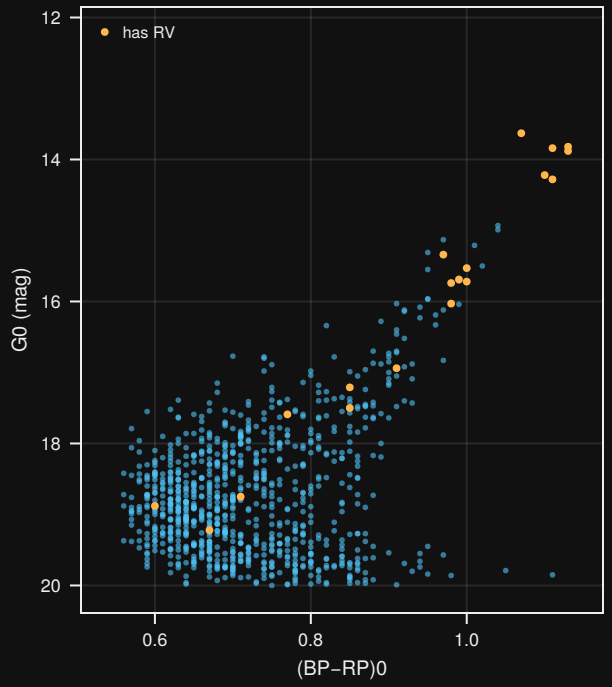




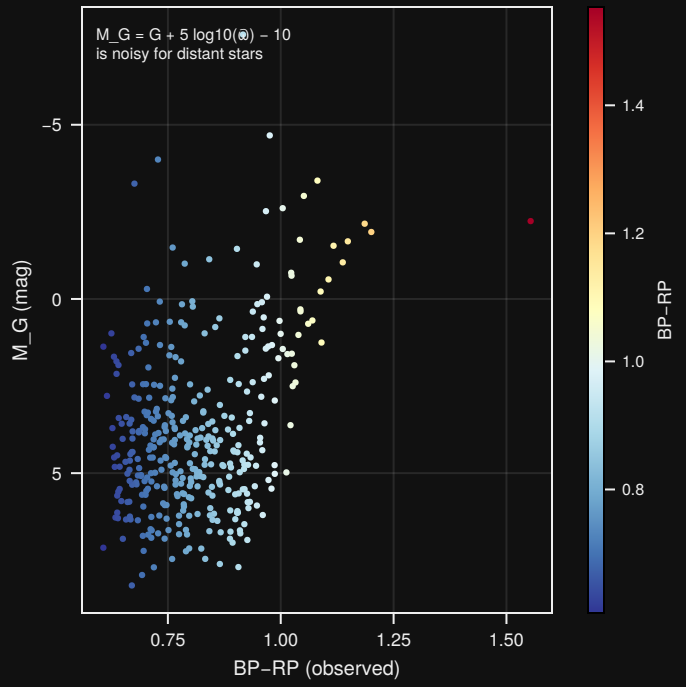


Ylgr

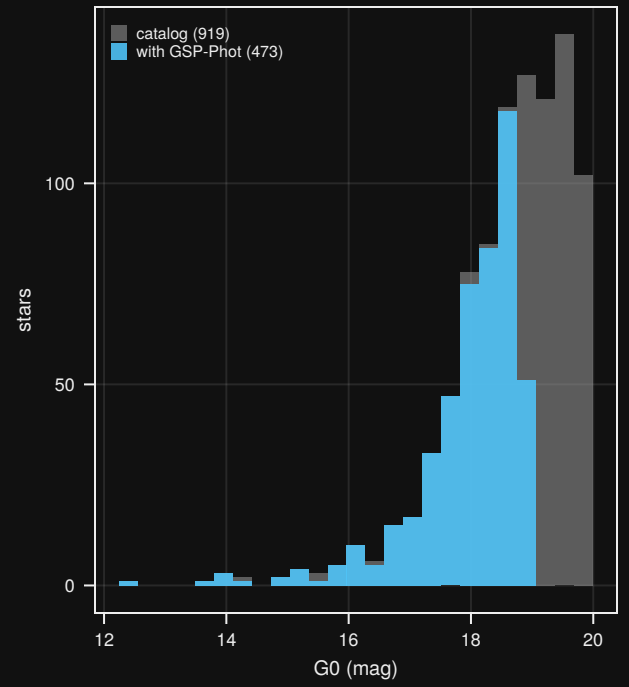
CMD · dereddened (n = 919)



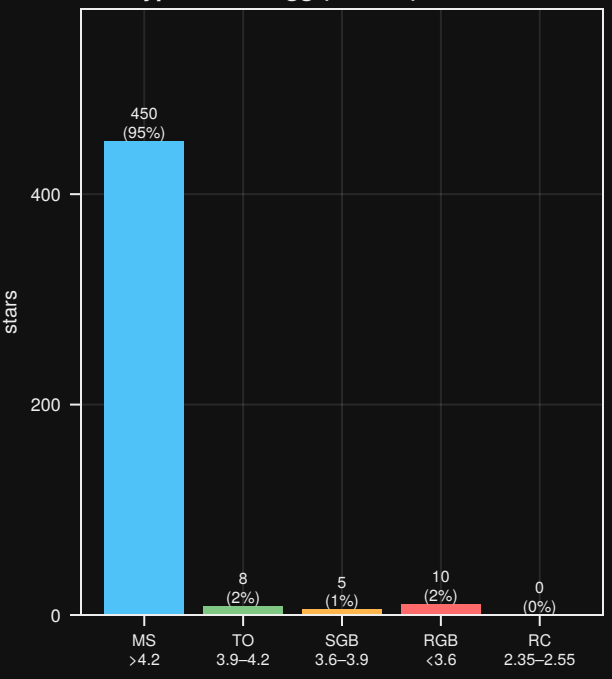
HR diagram · ϖ>0 (n = 371/473)



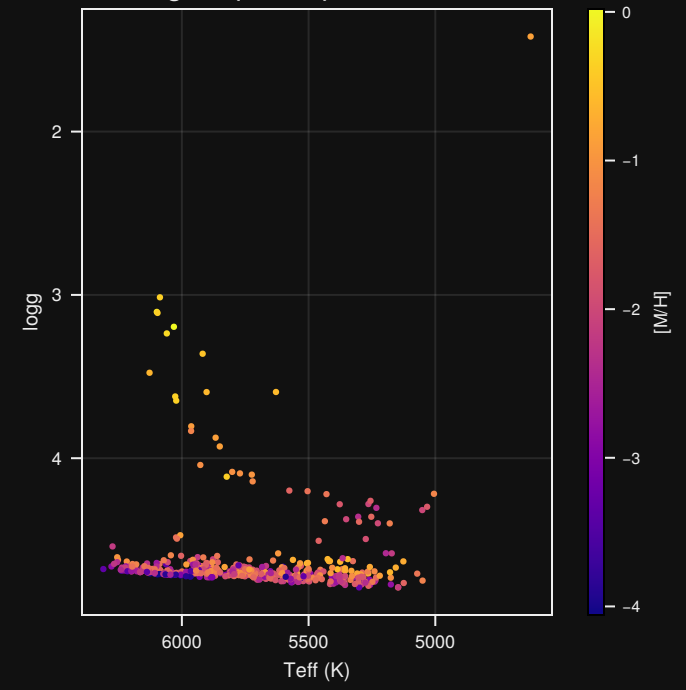
G0 & GSP-Phot coverage



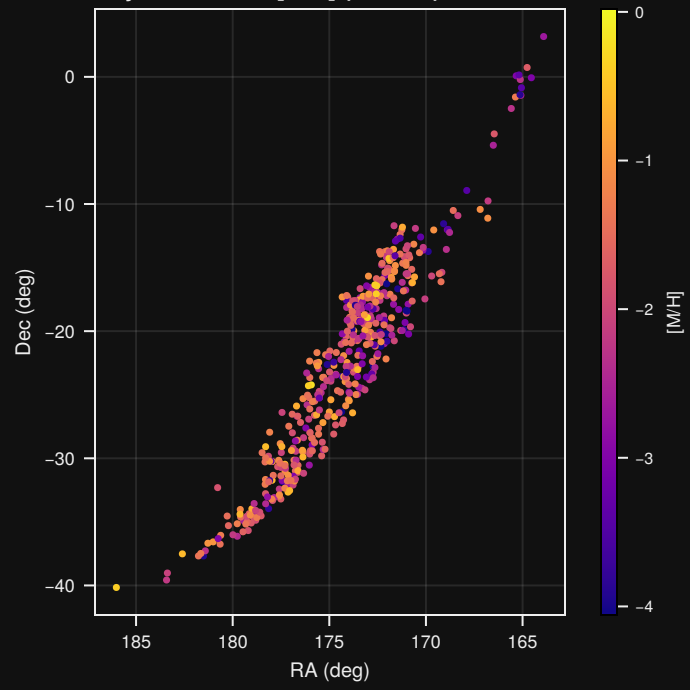
Star types from logg (n = 473)



Kiel diagram (n = 473)



Sky · GSP-Phot [M/H] (n = 473)

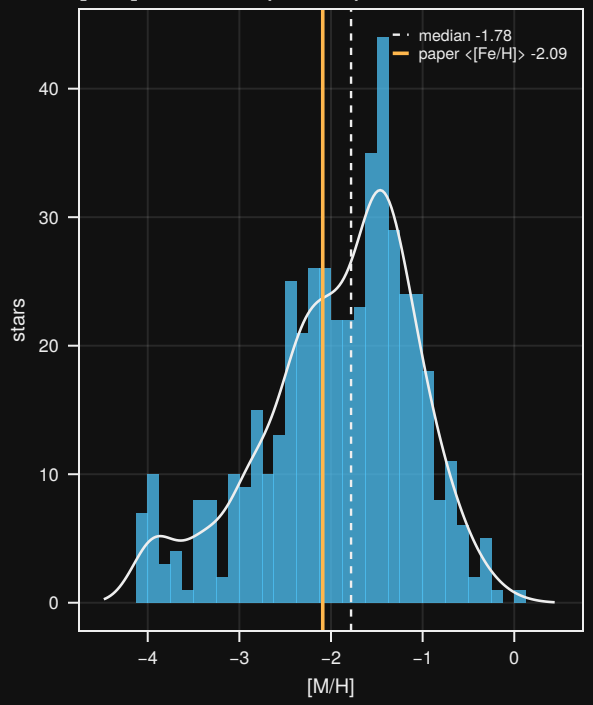


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

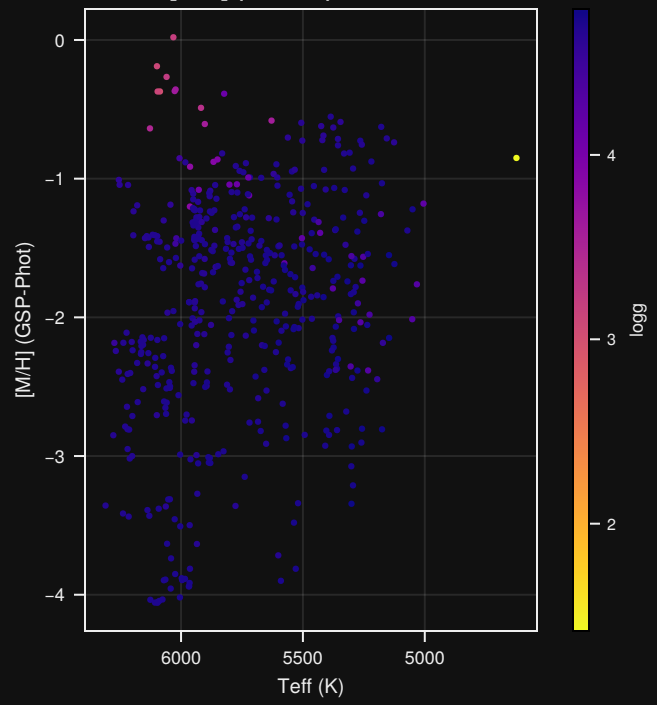
Ylgr

Metallicity · age · astrometric quality

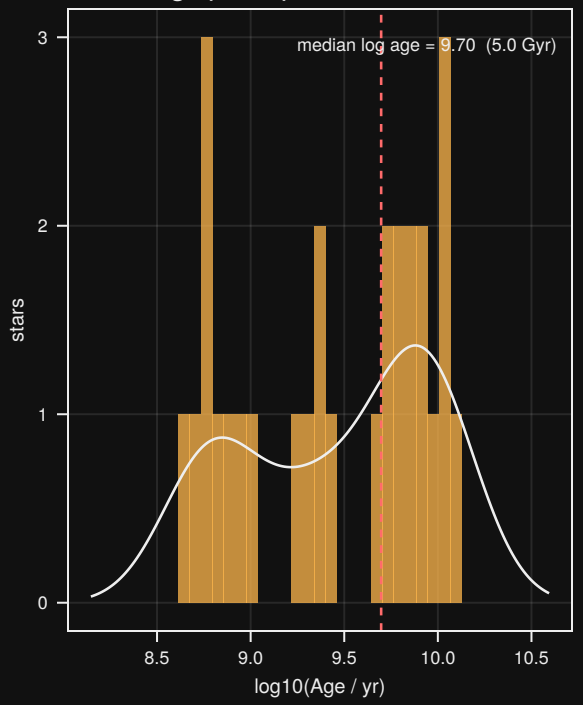
[M/H] GSP-Phot (n = 473)



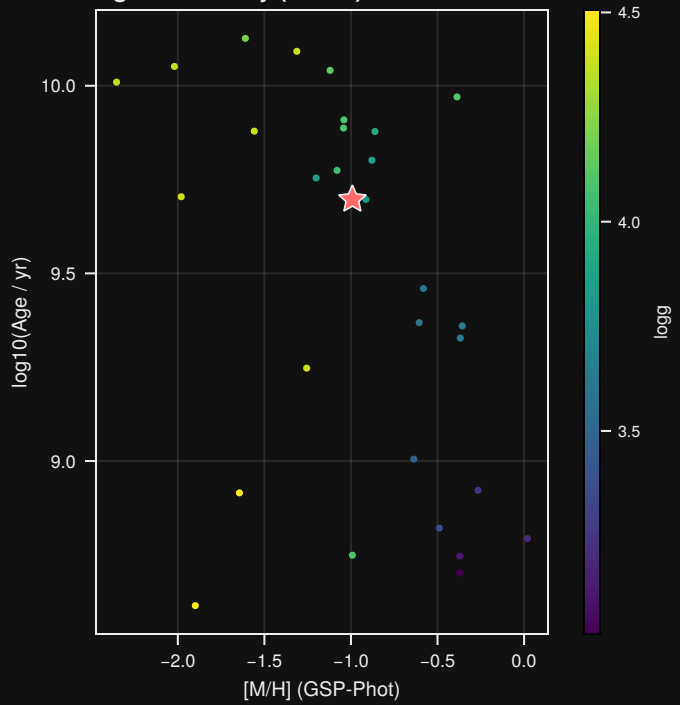
Teff vs [M/H] (n = 473)



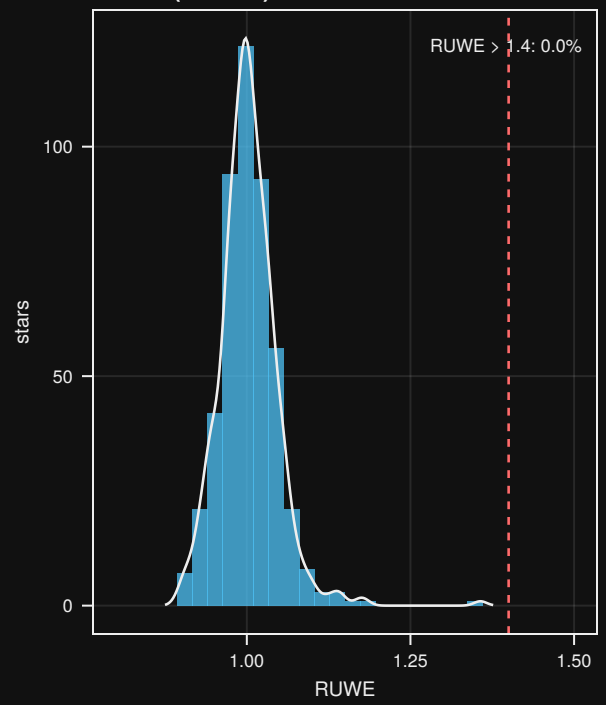
FLAME age (n = 29)



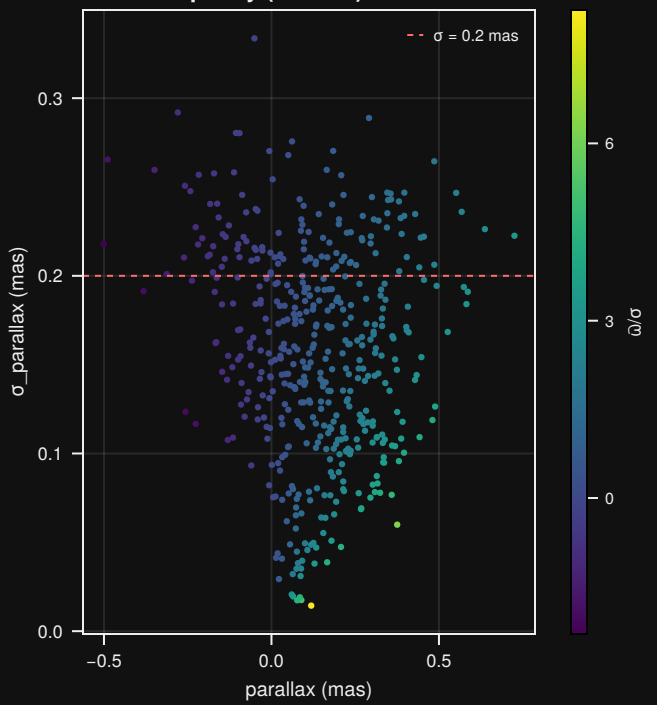
Age–metallicity (n = 29)



RUWE (n = 473)

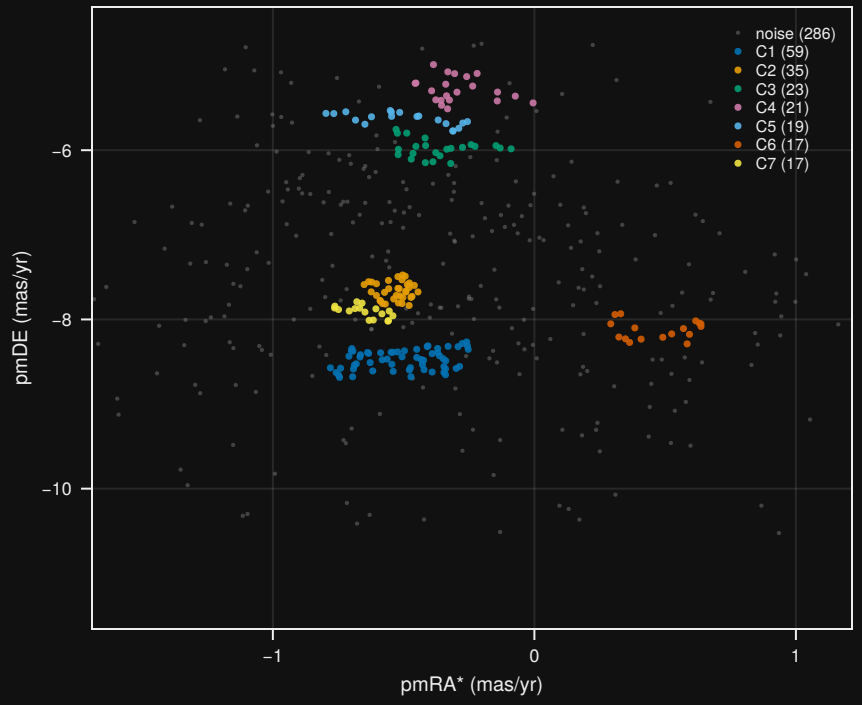


Parallax quality (n = 473)

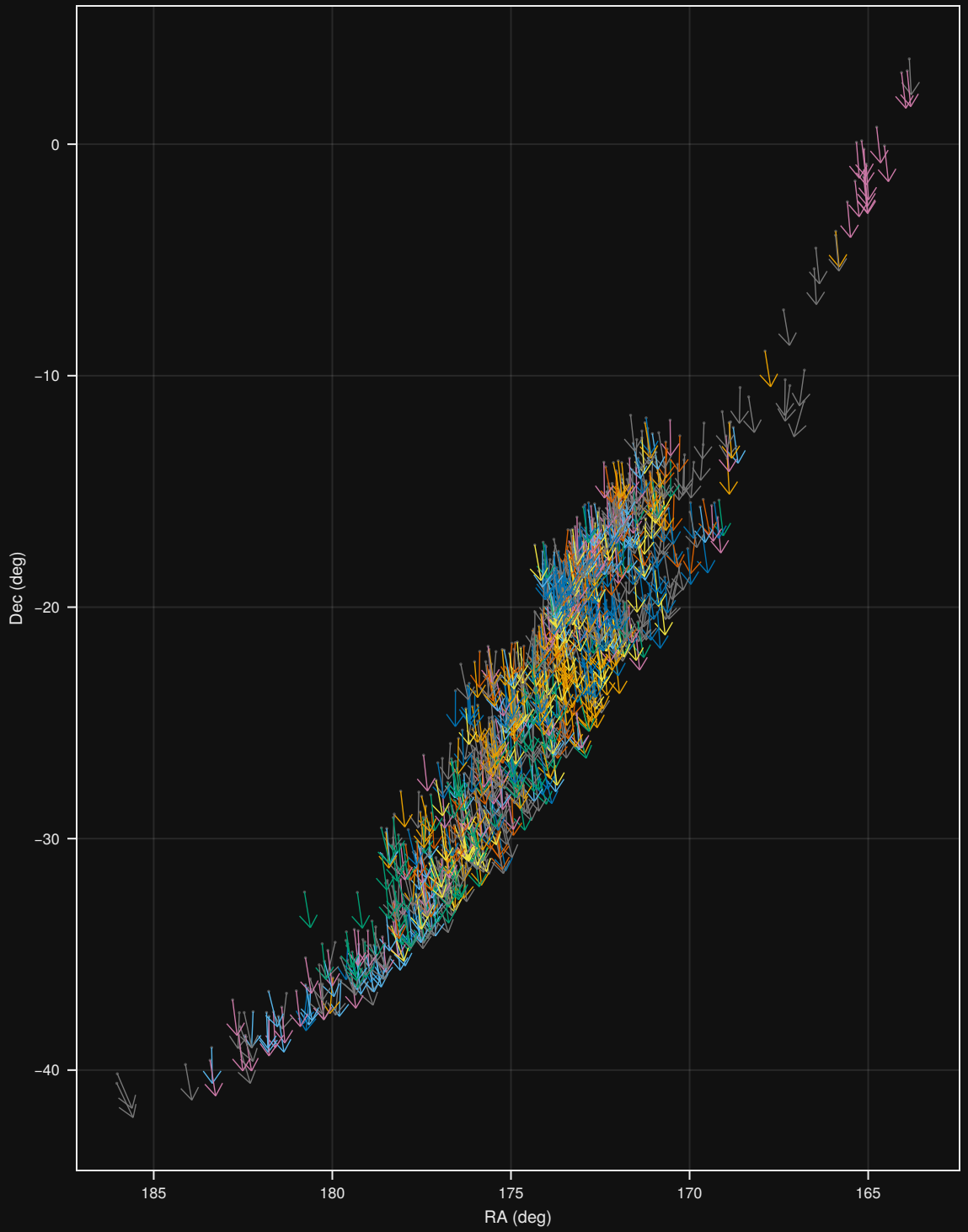


Ylgr

PM clustering · HDBSCAN · 60 cluster(s) · noise 31%



Proper-motion directions (arrows) coloured by cluster



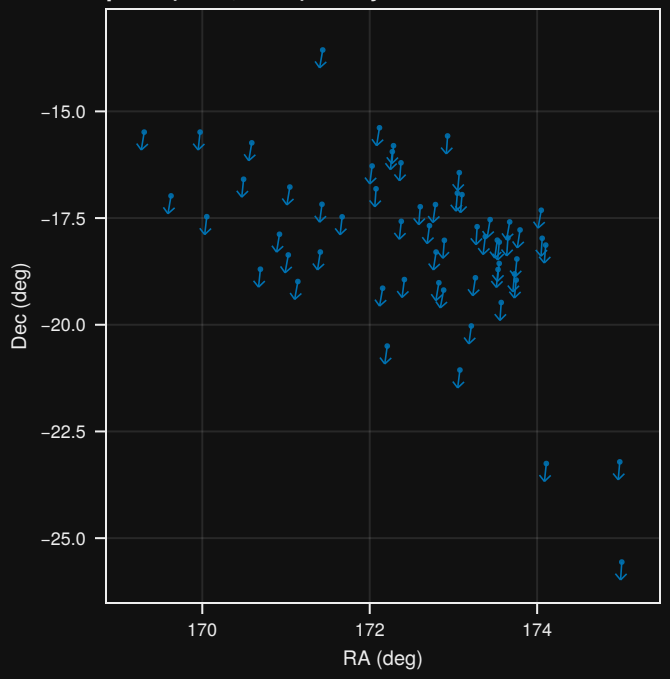
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	286	-0.33	-7.48	0.74	-1.58	4
C1	59	-0.48	-8.47	0.20	-2.20	0
C2	35	-0.52	-7.67	0.06	-1.74	3
C3	23	-0.39	-5.98	0.12	-1.64	1
C4	21	-0.33	-5.31	0.08	-1.44	1
C5	19	-0.52	-5.61	0.25	-1.76	0
C6	17	0.49	-8.11	0.19	-1.81	0
C7	17	-0.65	-7.90	0.09	-2.50	0
C8	17	-0.33	-8.06	0.05	-2.02	2

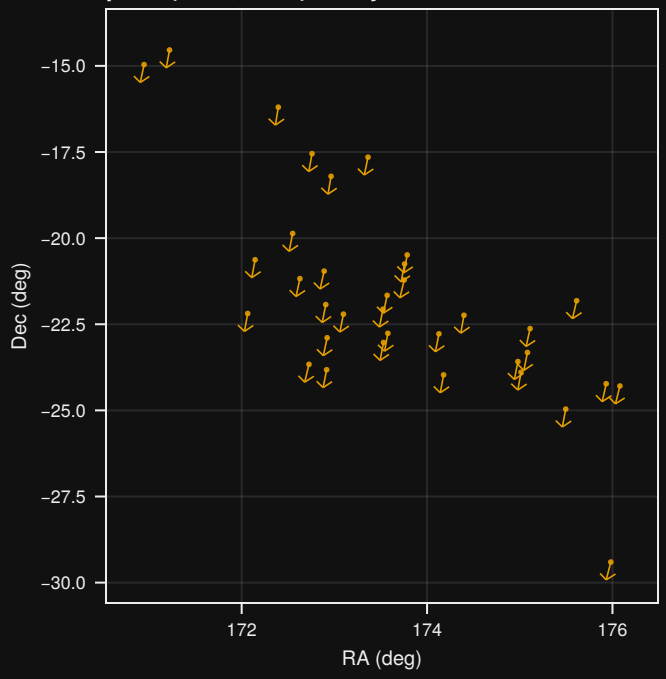
Ylgr

Per-cluster proper-motion vector maps

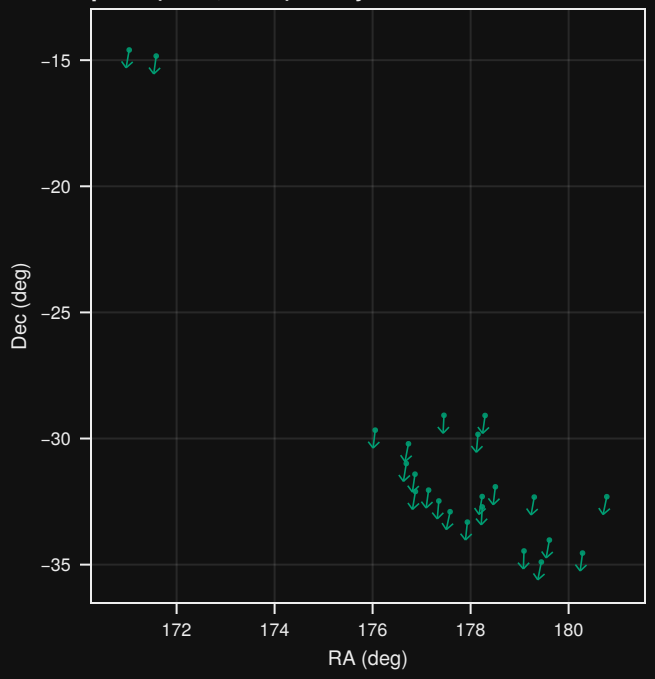
Cluster C1 · n = 59
pm = (-0.48, -8.47) mas/yr



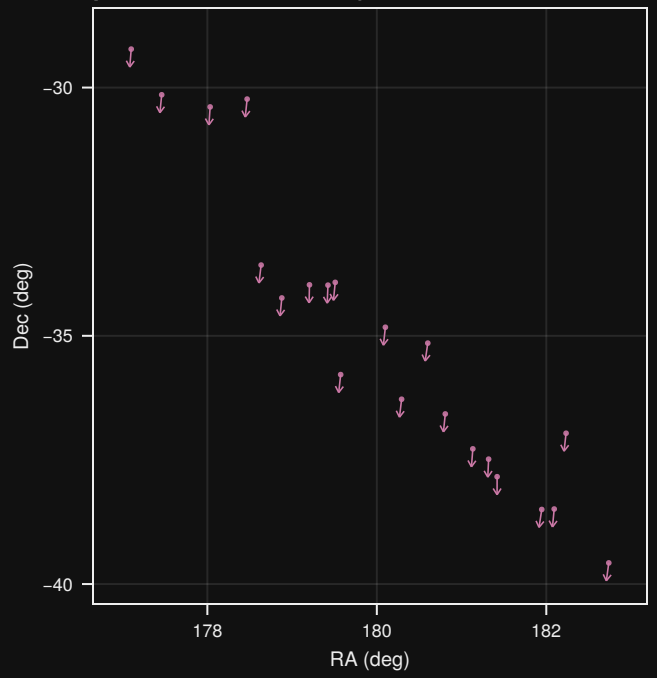
Cluster C2 · n = 35
pm = (-0.52, -7.67) mas/yr



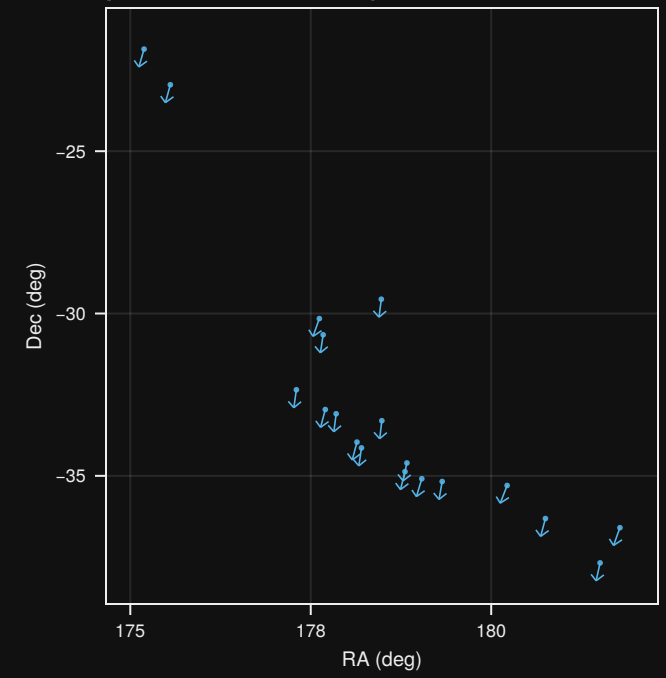
Cluster C3 · n = 23
pm = (-0.39, -5.98) mas/yr



Cluster C4 · n = 21
pm = (-0.33, -5.31) mas/yr



Cluster C5 · n = 19
pm = (-0.52, -5.61) mas/yr



Noise · n = 286

